Group 12 Phase 2 Report

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The Jupyter notebook and raw data in this presentation are accessible at https://github.com/siuzhuobin/Team12_IT5006_Healthcare_Analytics_AY2526.

We used ChatGPT5 as a reference on how to perform various data analysis tasks in Python and as a partial source of reference information. We are responsible for the content and quality of the submitted work.

Part 1: Problem Definition And Data Preparation

Problem Statement: Hospital readmission among patients with diabetes is common and costly. There are many different types of medications used to treat diabetes. A better understanding of how different treatments correlate with either increased or reduced risk of readmission will translate into safer treatment plans administered by physicians. To facilitate this, a machine learning model is constructed to track patient diabetic treatment regimen in real time to predict if the regimen leads to a hospital readmission.

The first step in preprocessing was data scrubbing. This involved removing null and irrelevant values, and exploring and addressing multicollinearity issues amongst the features. The columns listed in Table 1 were removed owing to their large numbers of null values.

Removed Data Columns (Null)	**Null Value Count**
weight	98569
max_glu_serum	96420
A1Cresult	84748
medical_specialty	49949
payer_code	40256

Table 1: Columns removed owing to large null count

We next removed the columns listed in Table 2, which comprise those that have only one unique data value and columns irrelevant to the problem statement.

Removed Data Columns (Irrelevant)	**Reason**

citoglipton	One value in column data
examide	One value in column data
encounter_id	Physician Case ID irrelevant for prediction
patient_nbr	Patient Number irrelevant for prediction

Table 2: Columns removed owing to irrelevance

Earlier EDA analysis has shown that the columns listed in Table 3 also had missing values. These columns were still utilized for training as the proportion of missing values was small and the missing values could be assigned to other values.

Modified Data Columns	**Proportion of NA Values**	**Modification**
race	2.2%	Converted NA to 'Other'
diag_1	1.4%	Converted ICD9 codes to integer values from floats. Codes that start with 'E' or 'V' were mapped to 1000 and those with unknown codes mapped to 0.
diag_2	0.4%	
diag_3	0.02%	

Table 3: Retained columns with missing values

The columns diag_1, diag_2, and diag_3 can be regarded as numerical data rather than categorical data since ICD9 codes have a large range from 0 - 999. The histogram frequency plots of these columns are shown in Fig. 1.

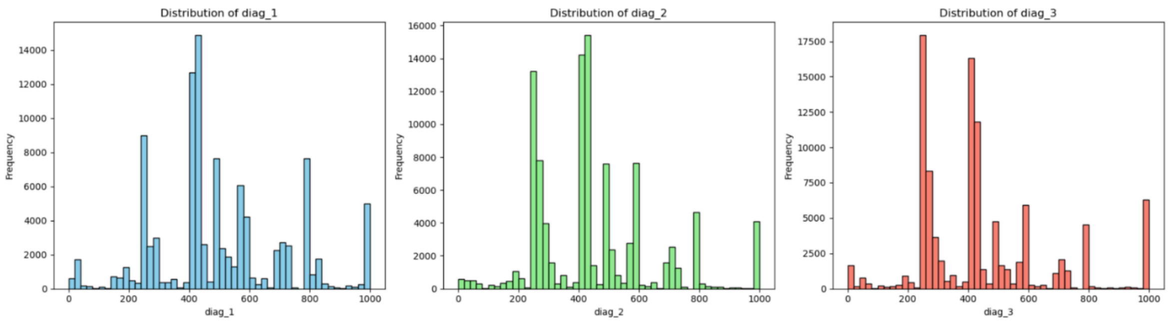

Figure 1: Histogram frequency plots of diag_1, diag_2, and diag_3

The next step after data scrubbing was data grouping and enhancement. Some categorical data were converted to numerical data while some numerous irrelevant categories in other categorical data were consolidated to an "other" category or merged with another category, as listed in Table 4.

Modified Data Columns	**Modification**
age	Convert age from intervals to mid-point numbers e.g '[10 20)' will be the numerical value 15
admission_type_id	1=Emergency, 2=Urgent,3=Elective, 4=other. Consolidate to other

discharge_disposition_id	1=Home, 3=SNF, 6=Home, 4=other Consolidate to other
admission_source_id	1=referral, 7=Emergency, 4=other Consolidate to other
Target Variable: readmitted	Merge ">30 days" and "No readmission" categories into a single "No readmission within 30 days" category. The main target variable will have two instead of three categories.

Table 4: Modified columns

Some new columns can be generated from current data to give a more holistic overview of the dataset, as listed in Table 5.

Added Columns	**Rationale For Addition**
num_drugs (Total number of drugs taken)	It is possible that the effect of an individual drug is less critical than the total drug load of treatment.
service_utilization (number_outpatient + number_emergency + number_inpatient)	Measures how much a patient utilizes medical services to gauge overall health.

Table 5: Added columns

Part 2: Model Implementation And Training

We tested three basic machine learning models comprising Logistic Regression, Random Forest, and Gradient Boosting on the cleaned data. The advantages and disadvantages of each model are presented in Table 6. We expect either the Random Forest or Gradient Boosting to give the highest accuracy, with the Random Forest model having the advantage with regard to overfitting. Tables 7 lists the obtained performance metrics for these models, which were first run without enhancements.

Machine Learning Model	**Advantages**	**Disadvantages**
Logistic Regression	Computationally Efficient	Can only learn linear decision boundaries. Low accuracy on data with complex, non-linear relationships.
Random Forest	Robust against overfitting High tolerance to null values and outliers High accuracy on data with complex non-linear relationships without requiring special feature engineering	High Memory Usage Computationally Intensive Blackbox Model, difficult to ascertain how the model learns on the data
Gradient Boosting	Achieves the highest predictive accuracy among the three, especially on structured data. Implementations, like XGBoost, can handle missing values intrinsically	Computationally Intensive Prone to overfitting Requires careful tuning of many hyperparameters Prone to outliers

Table 6: Comparison of Logistic Regression, Random Forest, and Gradient Boosting.

Logistic Regression

Readmission	Precision	Recall	F1 - Scores	ROC - AUC	PR - AUC
0 (No)	0.92	0.66	0.78	0.644	0.193
1 (Yes)	0.17	0.51	0.25		

Random Forest

Readmission	Precision	Recall	F1 - Scores	ROC - AUC	PR - AUC
0 (No)	0.89	1.00	0.94	0.637	0.195
1 (Yes)	0.50	0.01	0.01		

Gradient Boosting

Readmission	Precision	Recall	F1 - Scores	ROC - AUC	PR - AUC
0 (No)	0.89	1.00	0.94	0.680	0.226
1 (Yes)	0.57	0.01	0.02		

Table 7: Performance metrics for unenhanced models.

To see if we could improve the accuracy of our models, we performed the following additional enhancements:

Scaling Enhancement. The training data consists of both numerical data and categorical data. We first used the scaler.fit_transform on the numerical data, in which each numerical column was standardized by subtracting the mean and dividing by the standard deviation. This effectively scaled features to have a mean of 0 and standard deviation of 1. This step was a requirement for us to perform SMOTENC.

SMOTENC Enhancement. One concern with this dataset is that the imbalanced incidence of the target variable: Out of 101766 entries, 90409 of the target variables are NO and 11357 are YES. SMOTENC is a data augmentation technique to boost the target incidence with synthetic data.

For numerical data, SMOTENC generates artificial data by taking midpoints based on the Euclidean distance between real data. For categorical data, the new generated samples are decided by picking the most frequent category of the nearest neighbors present during the generation. We increased the YES values from 11357 to 90409 using SMOTENC. The obtained metrics are listed in Table 8.

Logistic Regression

Readmission	Precision	Recall	F1 - Scores	ROC - AUC	PR - AUC
0 (No)	0.90	0.82	0.86	0.584	0.169
1 (Yes)	0.17	0.29	0.21		

Random Forest

Readmission	Precision	Recall	F1 - Scores	ROC - AUC	PR - AUC
0 (No)	0.89	0.99	0.94	0.636	0.185
1 (Yes)	0.33	0.03	0.06		

Gradient Boosting

Readmission	Precision	Recall	F1 - Scores	ROC - AUC	PR - AUC
0 (No)	0.90	0.96	0.93	0.642	0.194
1 (Yes)	0.27	0.11	0.16		

Table 8: Performance metrics for models after scaling and SMOTENC.

Overall, we found that the use of SMOTENC was not satisfactory with a drop in ROC-AUC and a slight drop in precision for most cases. Two further enhancements were explored, without the use of SMOTENC:

StratifiedKFold Enhancement. Stratified K-Fold is an enhanced version of the standard K-Fold Cross-Validation for classification problems involving imbalanced datasets. When the normal K-Fold is used to perform a random split on an imbalanced dataset (e.g., 90% No and 10% Yes), the naive split will result in some folds having very few samples of the minority class (Yes). The evaluation is biased and unreliable, degrading model accuracy prediction on the minority class. Stratified K-Fold applies stratified sampling to ensure that each split will have the 9:1 ratio of No and Yes cases. Besides reducing bias because every set is guaranteed to have some minority data, stratified K-fold ensures that all observations appear in the trained and test datasets.

GridSearchCV Enhancement. GridSearchCV is used for hyperparameter tuning in machine learning. It systematically searches for the optimal combination of hyperparameters for a given model to maximize performance. It is not model specific and can be implemented across our choice of Logistic Regression, Random Forest and Gradient Boosting. A "grid" or dictionary of hyperparameters and a range of values to test for each model was first defined in order for GridSearchCV to subsequently evaluate the model for every possible combination of these values. The main disadvantage of this approach is that it is very computationally expensive. The most optimum model found may also be prone to overfitting. To address this, K-Fold or Stratified K-Fold is recommended before performing GridSearchCV.

Table 9 summarizes our findings after the GridSearchCV search was performed on the basic models to search for the best parameters. This was done without SMOTE. The best hyper parameters are listed in the second column with the corresponding best ROC - AUC score out of all the iterations carried out by the GridSearchCV search along with stratified K-Fold.

Model	**Best Hyperparameters (GridSearchCV)**	**Best ROC AUC**
Logistic Regression	clf__C: 2.0	0.640
Random Forest	clf__max_depth: 12	0.661

	clf__min_samples_leaf: 1 clf_n_estimators: 400	
Gradient Boosting	clf__learning_rate: 0.1 clf__max_depth: 3 clf_n_estimators:250	0.676

Table 9: Performance metrics for models after GridSearchCV and stratified K-fold

Of all of these models, Gradient Boosting has the highest performance because itutilizes careful parameter tuning for high accuracy. At the final stage of our experiment, we apply GridSearchCV to advanced variants of Gradient Boosting listed in Table 10.

Gradient Boosting Variant	**Advantages**	**Disadvantages**
XGB	Tolerate null values Optimized for parallel processing Built in cross-validation and regularization to reduce overfitting	Tuning complexity, has high number of parameters High computational load and memory usage
LGBM	Fastest and most memory efficient of the three Handle categorical data without one hot encoding	Prone to overfitting especially for small imbalanced datasets

Table 10: Advanced Gradient Boosting variants

To address the sensitivity of LGBM to imbalanced datasets, GridSearchCV was performed with either SMOTE or class weights to find the parameters with the best accuracy. SMOTE was used instead of SMOTENC because at this juncture we realized that SMOTENC may have been redundant because the categorical columns were already one-hot encoded.

Unlike SMOTE, synthetic data is not generated during class weights enhancement. Different levels of importance were assigned to each class during the training process according to the class weights. Higher weights are typically given to minority classes, while lower weights are assigned to majority classes. When the model makes an incorrect prediction on a sample from a highly-weighted (minority) class, the penalty (loss) incurred is greater than if it had made an error on a sample from a lower-weighted (majority) class. This encourages the model to pay more attention to correctly classifying the minority class. The results obtained are listed in Table 11.

Gradient Boosting Variant	Enhancement	Best Hyperparameters (GridSearchCV)	Best ROC AUC

XGB	Weighted	clf__colsample_bytree: 0.8 clf__learning_rate: 0.05 clf__max_depth: 4 clf__n_estimators: 300 clf__reg_lambda: 1.0 clf__scale_pos_weight: 5.970 clf__subsample: 0.8	0.676
LGBM	Weighted	clf__colsample_bytree: 0.8 clf__learning_rate: 0.05 clf__max_depth: 8 clf__n_estimators: 400 clf__num_leaves: 31 clf__reg_lambda: 1.0 clf__scale_pos_weight: 5.970 clf__subsample: 0.8	0.663
XGB	SMOTE	clf__colsample_bytree: 0.8 clf__learning_rate: 0.1 clf__max_depth: 4 clf__n_estimators: 600 clf__reg_lambda: 1.0 clf__subsample: 1.0	0.652
LGBM	SMOTE	clf__colsample_bytree: 0.8 clf__learning_rate: 0.05 clf__max_depth: -1 clf__n_estimators: 400 clf__num_leaves: 31 clf__reg_lambda: 1.0 clf__subsample: 0.8	0.663

Table 11: Best ROC AUC of optimized SMOTE- and class weight-enhanced XGB and LGBM

Part 3: Model Evaluation And Comparison

There are four basic prediction outcomes for the target variable, which takes the values of ‘Yes’ for readmission within 30 days and ‘No’ for no readmission within 30 days.

- True Positive (TP): Predict ‘Yes’, actually ‘Yes’
- False Positive (FP): Predicted ‘Yes’, actually ‘No’
- True Negative (TN): Predict ‘No’, actually ‘No’
- False Negative (FN): Predict ‘No’, actually ‘Yes’

A brief explanation of the metrics to measure model performance are given in Table 12.

Metrics	Explanation
Precision	TP / (TP + FP). - How accurately the positive label is predicted. If the model predicts most as negative, FP = 0 and one correct positive guess then precision will be 1.
Recall	TP / (TP + FN). - How accurately the positive label is found out of all positive labels. If the model predicts everything as positive, FN = 0 and recall will be 1.
F-1 Scores	2(Precision)(Recall) / (Recall + Precision) - The model may favor recall at the expense of prediction and vice versa. A good model maximizes both precision and recall. F1 score is the harmonic mean between precision and recall giving a good gauge of overall accuracy.
ROC - AUC	The ROC curve measures the tradeoff between the TP and FP rates. The higher the TP and the lower the FP for each threshold, the better the performance. In turn, this translates to a higher area under the ROC curve, which we measure.
PR - AUC	The PR curve measures the tradeoff between precision and recall. The higher the recall, the lower the precision. The PR AUC can be thought of as average precision scores calculated for each recall threshold. A good model will deliver simultaneously high levels of precision and high levels of recall. Like ROC, this translates to a higher area under the Precision Recall curve, which we measure.

Table 12: Performance metrics

PR - AUC examines the positive predictive value (PPV) and the true positive rate (TPR). Since PR AUC focuses mainly on the positive class (PPV and TPR), it cares less about the frequent negative class. We care more about the positive ‘Yes’ class because we have a highly imbalanced dataset where the fraction of positive classes, which we want to find (readmission cases), is small. Using PR AUC, which is more sensitive to the improvements for the positive class, is a better choice when there are no corrections (SMOTE / Class Weights) made to address imbalance.

Model (No SMOTE)	PR - AUC
Logistic Regression	0.193
Random Forest	0.195
Gradient Boost	0.226

Table 13: PR - AUC of basic models without SMOTE

We use PR - AUC to measure the performance of the model in our first test without any SMOTE or Class Weight enhancement. Gradient Boost is the best performing model, followed closely by Logistic Regression and Random Forest (Table 13). We note that despite data cleaning, PR - AUC accuracy is quite mediocre with all models achieving values below 0.25.

ROC - AUC should not be used for imbalanced datasets without correcting the imbalance either by using SMOTE or class weights. This is because for imbalanced datasets, the False Positive rate is pulled down by the high number of true negatives. If imbalance is corrected, ROC - AUC gives a better measure of model performance since it measures the predictions as a tradeoff between having many true positives and false positives at each threshold. A naive accuracy score simply collates how many correct predictions are made and does not take this tradeoff into account.

Model (With SMOTE)	ROC - AUC
Logistic Regression	0.585
Random Forest	0.636
Gradient Boost	0.642

Table 14: ROC - AUC of models after SMOTE

After we used SMOTE to generate synthetic data, we switched to ROC - AUC to measure the performance of the model. The results (Table 14) show that Gradient Boost is still the best performing model, followed closely by Logistic Regression and Gradient Boost.

Interestingly, we note that SMOTE actually decreased the ROC - AUC when applied to the dataset. Before SMOTE was applied, the ROC - AUC scores for the imbalanced data set were 0.644 (Logistic Regression), 0.637 (Random Forest), and 0.680 (Gradient Boost). This contradicts our expectation that the ROC - AUC scores would be improved by SMOTE.

Advance Gradient Boost Models (With GridSearchCV Tuning)	Best Possible ROC - AUC
XGB (Weighted Class)	0.673
LGBM (Weighted Class)	0.666

Table 15: ROC - AUC of advanced Gradient Boost variants

Lastly we examine the best possible ROC - AUC scores for the advanced Gradient Boost variants in the SMOTE and Weighted Class cases (Table 15). We note that the Weighted Class enhancement all produce marginally better ROC - AUC values than the SMOTE enhancement.

For all cases, we never hit a ROC - AUC score of 0.7 and above. This is true even after automated parameter tuning to find the parameters that give the best accuracy. From these observations, we think that there might be deeper and more complex patterns in the dataset that require deep learning to fully capture. Secondly, the use of SMOTE and SMOTENC did not improve model accuracy and in some cases, degraded it. This suggests that SMOTE actually generated noisy or unrealistic synthetic samples near class boundaries and is not appropriate for the training data.

Part 4: Model Interpretation and Business Insights for 30-Day Readmission

Based on the results of the weighted XGBoost model and the SHAP analysis performed, the following top 5 features stood out:

1. **Number of inpatient visits in year preceding encounter**
   A higher number of inpatient visits in the preceding year predicts a higher risk of 30-day readmission. This is consistent with what has been observed in literature (Brennan et al., 2015). Qualitatively, a reason for this could be that patients with more complex medical issues would need to be readmitted more often. Higher readmission rates in the past would then signal a higher likelihood that a patient would readmit.
2. **Discharge disposition**
   Non-routine discharge dispositions that imply ongoing clinical needs after hospitalization, such as to nursing homes or intermediate care, tend to push risk upwards. This aligns with transitions-of-care evidence for inadequate discharge arrangements or inadequately safe and seamless transitions of care leading to hospital readmissions. (Earl et al., 2020)
3. **Primary diagnoses**
   The main primary diagnosis plays an important factor in predicting if a patient is going to be readmitted soon. This makes sense because patients with chronic diseases like cardiovascular, respiratory, and endocrine-related diseases have a higher chance of readmission. Mapping codes to broad clinical groupings helps the model surface these patterns and match clinical guidelines that certain diagnoses present high risks for readmissions (Kansagara et al., 2011).
4. **Length of hospital stay**
   Consistent with our literature review, the length of hospital stay ranks highly among the important features for predicting readmission. Although its predictive power in our models is not as high as we would have expected from the literature review, it makes qualitative sense that length of hospital stay would still be an important feature as a longer hospital stay signals higher disease or even social complexity (Rachoin et al., 2020).
5. **Number of diagnoses**
   Again, consistent with our literature review, the presence of comorbidities positively predicts readmission, though not as strongly as we expected. Nevertheless, it still makes qualitative sense for comorbidities to increase risk of readmission as having multiple diseases signal the need for more complex medical care (Mukhopadhyay et al., 2021).

The weighted XGB and LGBM models also scored higher recalls as compared to precision. This is fine because in critical settings such as healthcare where missing a true readmission case would be highly detrimental to both the hospital and patient, we would prefer to have false positives as opposed to false negatives. However, a caveat to take note is that flagging too many false positives may present an operational strain to the healthcare system. It is worthwhile to fine-tune and refine our models even further to improve their higher predictive power.

Based on the results, our approach can be used operationally in hospitals to detect potential readmission cases at the time of discharge, or near the time of discharge. The model can be a useful triaging tool to flag a patient to the respective care team so medical interventions can be performed or the patient's discharge and post-hospitalization care plan reconsidered carefully. Patients who are at higher risk of readmission can have earlier follow-ups, higher social support, and telehealth check ins. However, as the predicting power of our model is still modest, it cannot be used as a fully automated tool to automatically select patients and modify their care plan and human intervention is still necessary. The model can be used to prioritize patients, but ultimately their care plan would still need to be individualized by their doctors who have domain expertise and are guided by clinical guidelines.

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